

Chapter 5 - The Korteweg-deVries Equation - Notes

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1 The Korteweg-deVries Equation

In 1834, J.S. Russell observed the phenomena of a large bulge of water slowly traveling along a channel of water. The ability of this water wave to retain its shape for such a long period of time was quite remarkable and led Russell to study this disturbance by conducting numerous detailed experiments. He later came to call this phenomena a Wave of Translation, highly suggestive of a traveling wave. Russell's work on the Wave of Translation is now considered the beginning study of what are now called solitary waves or solitons.

1.1 The KdV Equation

The **Korteweg-deVries** (KdV) equation is a PDE used to model the height of the surface of shallow water in the presence of long gravity waves. In these waves, the length of the wave is large compared to the depth of the water.

$$U_t + (a_1 + a_2 U)U_x + a_3 U_{xxx} = 0, \quad a_2, a_3 \neq 0 \quad (1)$$

It is a third order non-linear equation. Scaling x and t , and using a substitution $u = a_1 + a_2 U$ yields the reduced, more commonly used form of the KdV equation.

$$u_t + uu_x + u_{xxx} = 0 \quad (2)$$

1.2 Solitary Wave Solutions

The solutions found for KdV are called **solitons** and model the wave phenomenon observed by Russel. We can look at the traveling wave solution in the form of a pulse.

$$u(x, t) = f(x - ct) \text{ where } c > 0 \text{ and } f^n(z) \rightarrow 0 \text{ as } z \rightarrow \pm\infty$$

The first step is to plug in our derivatives of f into our KdV equation.

$$-cf' + ff' + f''' = 0$$

Then we integrate once (*Hint: chain rule*).

$$-cf + \frac{1}{2}f^2 + f'' = a$$

Since we know $f(z)$ and $f''(z) \rightarrow 0$ as $z \rightarrow \infty$, the value of a is 0. We use the trick of multiplying by f' to integrate again.

Multiply by f' :

$$-cf f' + \frac{1}{2}f^2 f' + f' f'' = 0$$

Integrate :

$$-\frac{1}{2}cf^2 + \frac{1}{6}f^3 + \frac{1}{2}(f')^2 = b$$

Since we know $f(z)$ and $f''(z) \rightarrow 0$ as $z \rightarrow \infty$ (form of pulse), the value of b is 0. We can now solve for $(f')^2$.

$$\begin{aligned} -\frac{1}{2}cf^2 + \frac{1}{6}f^3 + \frac{1}{2}(f')^2 &= 0 \\ -3cf^2 + f^3 + 3(f')^2 &= 0 \\ 3(f')^2 &= 3cf^2 - f^3 \\ 3(f')^2 &= (3c - f)f^2 \end{aligned}$$

Now from here, we need to keep in mind that $0 < f(z) < 3c$ in order to have a positive RHS (because we need to take a square root).

$$\begin{aligned}
3(f')^2 &= (3c - f)f^2 \\
\sqrt{3(f')^2} &= \sqrt{(3c - f)f^2} \\
\sqrt{3}f' &= \sqrt{(3c - f)}f \\
\frac{\sqrt{3}}{\sqrt{(3c - f)}f}f' &= 1
\end{aligned}$$

To integrate the LHS, we can make the substitution $g^2 = 3c - f$, with $f = 3c - g^2$ and $f' = -2gg'$

$$\begin{aligned}
\frac{\sqrt{3}}{\sqrt{(3c - f)}f}f' &= 1 \\
\frac{\sqrt{3}}{\sqrt{g^2(3c - g^2)}}(-2gg') &= 1 \\
\frac{\sqrt{3}}{\sqrt{g^2(3c - g^2)}}(2gg') &= -1 \\
\frac{2gg'\sqrt{3}}{\sqrt{g^2(3c - g^2)}} &= -1 \\
\frac{2gg'\sqrt{3}}{g(3c - g^2)} &= -1 \\
\frac{2g'\sqrt{3}}{(3c - g^2)} &= -1 \\
\frac{2\sqrt{3}}{(3c - g^2)}g' &= -1
\end{aligned}$$

Now, we can integrate with respect to z , using partial fractions.

$$\int \frac{2\sqrt{3}}{3c - g^2} g' dz = \int -1 dz$$

$$\int \frac{2\sqrt{3}}{3c - g^2} dg = -z + C$$

Apply partial fractions: $\frac{2\sqrt{3}}{(\sqrt{3c} - g)(\sqrt{3c} + g)} = \frac{A}{\sqrt{3c} - g} + \frac{B}{\sqrt{3c} + g}$

$$2\sqrt{3} = A(\sqrt{3c} + g) + B(\sqrt{3c} - g)$$

Set $g = \sqrt{3c}$: $2\sqrt{3} = A(2\sqrt{3c}) \implies A = \frac{1}{\sqrt{c}}$

Set $g = -\sqrt{3c}$: $2\sqrt{3} = B(2\sqrt{3c}) \implies B = \frac{1}{\sqrt{c}}$

$$\int \left(\frac{1/\sqrt{c}}{\sqrt{3c} - g} + \frac{1/\sqrt{c}}{\sqrt{3c} + g} \right) dg = -z + C$$

$$\frac{1}{\sqrt{c}} \left(-\ln(\sqrt{3c} - g) + \ln(\sqrt{3c} + g) \right) = -z + C$$

$$\frac{1}{\sqrt{c}} \ln \left(\frac{\sqrt{3c} + g}{\sqrt{3c} - g} \right) = -z + C$$

$$\ln \left(\frac{\sqrt{3c} + g}{\sqrt{3c} - g} \right) = -\sqrt{c}z + C\sqrt{c}$$

Let $d = C\sqrt{c}$: $\ln \left(\frac{\sqrt{3c} + g}{\sqrt{3c} - g} \right) = -\sqrt{c}z + d$

Now we can solve for $g(z)$:

$$\ln \left(\frac{\sqrt{3c} + g}{\sqrt{3c} - g} \right) = -\sqrt{c}z + d$$

Exponentiate both sides: $\frac{\sqrt{3c} + g}{\sqrt{3c} - g} = e^{-\sqrt{c}z + d}$

Multiply by the denominator: $\sqrt{3c} + g = (\sqrt{3c} - g)e^{-\sqrt{c}z + d}$

Distribute the right side: $\sqrt{3c} + g = \sqrt{3c}e^{-\sqrt{c}z + d} - ge^{-\sqrt{c}z + d}$

Group g terms on the left: $g + ge^{-\sqrt{c}z + d} = \sqrt{3c}e^{-\sqrt{c}z + d} - \sqrt{3c}$

Factor out g and $\sqrt{3c}$: $g(1 + e^{-\sqrt{c}z + d}) = \sqrt{3c}(e^{-\sqrt{c}z + d} - 1)$

Divide to isolate $g(z)$: $g(z) = \sqrt{3c} \frac{e^{-\sqrt{c}z + d} - 1}{e^{-\sqrt{c}z + d} + 1}$

$$\sqrt{3c} \frac{e^{-\sqrt{c}z + d} - 1}{e^{-\sqrt{c}z + d} + 1} = \sqrt{3c} \tanh \frac{1}{2}(\sqrt{c}z - d)$$

Now using our substitution $f = 3c - g^2$

$$f(z) = 3c \operatorname{sech}^2 \left(\frac{1}{2}(\sqrt{c}z - d) \right)$$

And d is just a shift of the shape so

$$f(z) = 3c \operatorname{sech}^2 \left(\frac{1}{2}(\sqrt{c}z) \right)$$

So the resulting traveling wave solution to the KdV equation is ($z = x - ct$)

$$u(x, t) = 3c \operatorname{sech}^2 \left(\frac{1}{2}(\sqrt{c}(x - ct)) \right)$$

$$u(x, t) = 3c \operatorname{sech}^2 \left(\frac{\sqrt{c}}{2}(x - ct) \right)$$

This solution is called a **solitary wave** or **soliton** solution.